

Research & Report (Theory Task)

1. Types of malware and their impact.

Malware (Malicious Software) is a harmful program designed to damage, steal or control a system without permission. Hackers use malware to steal data, spy, or demand ransom from users.

1. Virus

A program that attaches to files/software and spreads when opened.

Impact : Corrupts files, slows down the system and can crash computers.

2. Worms

A self-spreading malware that does not need user action.

Impact : Overload networks, spreads to other systems and slows down devices.

3. Trojan Horse

A fake program that looks useful but gives hackers control over a system.

Impact : Hackers steal data, delete files or spy on users.

4. Ransomware

A dangerous malware that locks files and demands ransom to unlock them.

Impact : Hackers steal sensitive information like bank details.

5. Spyware

Secretly tracks keystrokes, passwords and browsing history.

Impact : Hackers steal sensitive information like bank details.

2, Famous Malware Attacks

1. WannaCry Ransomware (2017)

What happened?

->In May 2017, WannaCry ransomware **attacked 200,000+ computers in 150+ countries** using a Windows security vulnerability. It affected hospitals, banks and major institutions worldwide.

How Did it Work?

Used a **Windows flaw** (EternalBlue exploit), original leaked from the **NSA** (USA's National Security Agency).

Once inside, it encrypted all files and showed a ransom message demanding **Bitcoin payment**

If the ransom wasn't paid, the files were permanently deleted.

Impact :

1. Hospitals, banks, and railway networks suffered losses.
2. The UK's National Health Service (NHS) was badly affected.

How Was It Stopped?

Microsoft released a security patch to fix the vulnerability.

A cybersecurity expert found a **kill switch** that stopped its spread.

2. Stuxnet – The Cyber Weapon (2010)

What Happened?

->Stuxnet was a highly advanced cyber weapon discovered in 2010, believed to be created by the **USA & Israel** to sabotage Iran's **nuclear program**.

How Did It Work?

Targeted Siemens industrial control systems in Iran's nuclear plants.

Secretly hacked uranium centrifuges, making them spin too fast or too slow, **damaging them permanently**.

Iran's nuclear program was set back by years.

Impact :

1. **First cyber weapon** used to destroy physical infrastructure.
2. **Started a new era of cyber warfare.**
3. Countries began investing in **cyber defense & offensive cyber tools**.

3. Pegasus Spyware – The Ultimate Phone Hacker

What is Pegasus?

->Pegasus is a **highly advanced spyware** developed by **NSO Group (Israel)** that can hack **Android & iPhone** devices **without users knowing**.

How Does it Work?

Spreads via malicious links (SMS, WhatsApp, Email) or even missed WhatsApp calls!

Once installed, it can:

1. **Read encrypted messages** (WhatsApp, Signal)
2. **Record calls & track location**
3. **Control camera & microphone**

Famous Incidents :

- 1) **2019 WhatsApp attack** – 1400 journalists, activists and politicians were hacked.
- 2) **2021 Pegasus Project Leak** - World leader's phones were targeted.

Impact :

1. Huge privacy & security concerns worldwide.
2. Governments were accused of spying on journalists & political opponents.
3. Apple & WhatsApp sued NSO Group for misusing their platform.